

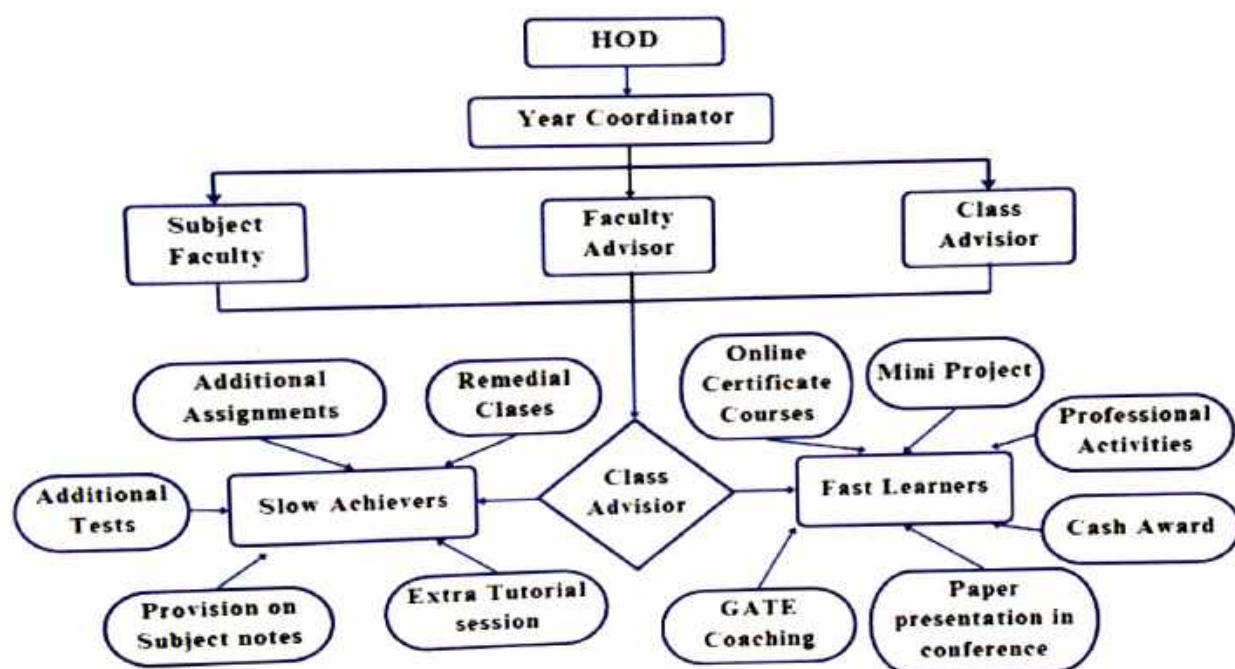


S.A ENGINEERING COLLEGE, CHENNAI - 77

(Autonomous - Institute Level Research Centre, Affiliated to Anna University)

Department of Computer Science and Engineering

PROCESS TO ANALYZE WEAK AND BRIGHT STUDENTS



P9. Records of list of methodologies used to support weak students and encourage bright students and impact analysis.

Student Categorization and Identification Methods

- **Weak Students (Slow Learners):** Identified based on low performance in internal assessments (scoring below 50%). Faculty advisors monitor progress through mentoring sessions.
- **Bright Students (Fast Learners):** Identified based on high performance in internal assessments (scoring above 75%). Encouraged to explore advanced topics and participate in competitions.

S.No.	Methodology	Description / Usage	Impact / Outcome
1	Remedial Classes	Extra classes with paced teaching, simplified course materials, practical examples, and evening doubt-clearing sessions to improve performance.	Improved subject understanding and pass percentage.
2	Additional Assignments	Personalized and supplementary assignments to bridge learning gaps and enhance understanding.	Reinforced core concepts and better internal assessment scores.
3	Additional Tests	Conducting retests and internal assessments to evaluate improvement and strengthen exam preparedness.	Enhanced confidence and exam readiness.
4	Provision of Subject Notes	Simplified lecture notes and learning materials provided for better comprehension and self-study.	Improved self-learning and concept clarity.
5	Extra Tutorial Sessions	Additional tutorial hours and one-on-one mentoring for concept clarity.	Better academic performance through personal attention.
6	Counselling & Motivation	Faculty advisors counsel students to overcome academic stress, improve focus, and build confidence.	Positive attitude and increased academic motivation.
7	Parent-Teacher Meeting	Regular meetings organized to discuss student progress, attendance, and academic performance with parents for collaborative support.	Strengthened communication between parents and faculty leading to improved student discipline, performance, and engagement.

Table 1: Methodologies Adopted to Support Weak Students

S.No.	Activity / Methodology	Description / Usage	Impact / Outcome
1	Online Certificate Courses (NPTEL / Coursera / Infosys Springboard)	Students are encouraged to enroll in certified online courses to gain exposure to cutting-edge technologies and supplement their academic curriculum.	Enhanced technical competency, domain knowledge, and self-learning ability.
2	Mini Projects	Implementation of project-based learning where students develop small innovative projects under faculty guidance to apply theoretical concepts practically.	Improved problem-solving, creativity, and hands-on implementation skills.
3	GATE Coaching / GATE Classes	Organized coaching sessions and mock tests for GATE and other national-level competitive examinations like CAT, GRE, etc.	Increased number of students qualifying in competitive exams and pursuing higher studies.
4	Paper Presentation in Conferences	Students are motivated and guided to prepare and present technical papers in national and international conferences or journals.	Enhanced research aptitude, technical documentation, and presentation skills.
5	Professional Activities (IEEE)	Encouraging participation in professional bodies, technical societies, and student chapters to develop technical and managerial exposure.	Strengthened professional networking, leadership qualities, and collaborative learning.
6	Cash Awards & Recognition	Awards, certificates, and medals are presented to students for outstanding academic, research, and extracurricular achievements.	Promoted healthy competition, motivation, and pursuit of excellence.
7	Value Added Course by Foreign resource person / Industry Experts	Short-term credit courses delivered by foreign professors or industry experts offering specialized training beyond the regular curriculum.	Broadened global perspective and awareness of advanced industrial trends.
8	Japanese Language Classes	Offering foreign language training to students for improving global communication and employability skills.	Improved linguistic competency and placement opportunities in multinational organizations.
9	Industry Training Courses (WIPRO, CISCO, AWS, etc.)	Collaboration with industry partners to offer certification programs and hands-on training in trending technologies.	Enhanced industry readiness, practical exposure, and employability skills.
10	Club Activities (Technical / Coding / Innovation / Cultural Clubs)	Formation of student clubs such as Coding Club, Innovation Club, Green Club, and Cultural Club to promote extracurricular learning and creativity through events, workshops, and hackathons.	Developed teamwork, leadership, innovation, and communication skills among students.
11	On-Duty Training	Students are given on-duty	Improved placement

	for Placement (2 Hours per day)	permission for attending placement training sessions including aptitude, communication, and technical skill enhancement classes.	preparedness, communication skills, and employability outcomes.
12	Internship On-Duty (Online Internship)	On-duty facility is provided for students undertaking approved online internships in collaboration with industries or platforms.	Facilitated practical exposure, professional experience, and strengthened industry-academia linkage.

Table 2: Methodologies Adopted to Support Bright Students

3. Impact Analysis

For Weak Students:

- Improved understanding of fundamental concepts and problem-solving skills.
- Better performance in internal and semester-end examinations.
- Enhanced confidence, attendance, and participation in class activities.
- Strengthened programming, analytical, and communication abilities.

For Bright Students:

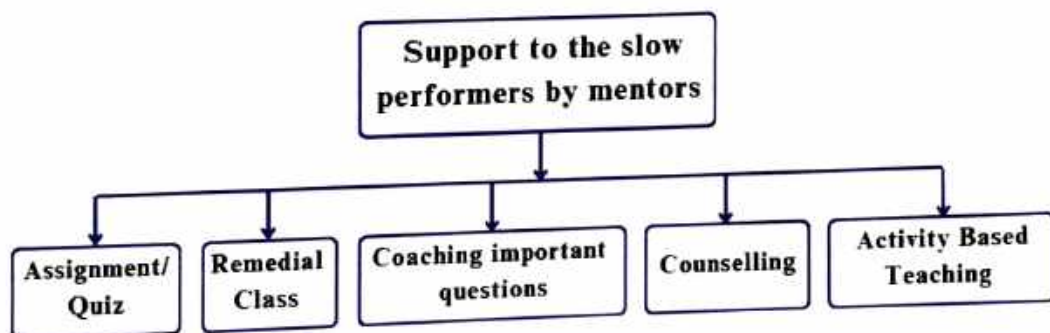
- Encouragement in professional growth through workshops, hackathons, and online courses.
- Exposure to research, publications, and technical competitions.
- Recognition through awards, certificates, and prizes motivating further excellence.
- Development of teamwork, leadership, and innovation skills.



S.A ENGINEERING COLLEGE, CHENNAI - 77

(Autonomous - Institute Level Research Centre, Affiliated to Anna University)
Department of Computer Science and Engineering

SUPPORT TO THE SLOW LEARNERS



OBJECT ORIENTED

PROGRAMMING - I A

§

R.No: 2416196

Reg No: 1119242501074

Part - A

Answer all the questions

Four Java Buzzwords:

- * Simple
- * Object Oriented
- * Platform Independent
- * Secure.

Encapsulation:

Encapsulation is the process of binding data (variables) and methods into a single unit (class) and restricting direct access to data using access modifiers.

Ex:

```
class Student
{
    private int age;
    public void setAge (int a)
    {
        age = a;
    }
    public int getAge ()
    {
        return age;
    }
}
```

3. Uses of constructor:

- * Initializes object automatically when created
- * Eliminates need of separate initialization method
- * Supports constructor overloading
- * Enhances code readability and reusability.

4. Package is created and used in Java:

Creation: `package mypack;` written at the top of the file.

Save file as `MyClass.java` inside folder `my`

Import `mypack.MyClass;`

5. Java support multiple Inheritance:

No Java does not support multiple inheritance using classes to avoid ambiguity problem diamond problem.

But multiple inheritance is supported using interfaces.

6. Use of the final keyword:

Uses:

final variable → constant

final method → cannot be overridden

final class → cannot be inherited


```
final int MAX = 100;  
final class Car { }
```

data is written to a file:

```
import java.io.*;
```

```
class FileExample {
```

```
    public static void main (String[] args) throws  
                                Exception
```

```
{  
    FileWriter fw = new FileWriter ("test.txt");
```

```
    fw.write ("Hello Java!");
```

```
    fw.close ();
```

```
}
```

Two methods of the Input Stream class:

int read () → reads one byte

void close () → closes input stream

Use of default exception handler:

If no try-catch is used, Java's default exception handler handles the exception.

It prints the exception name, description and stack trace.

10. Catch and finally block:

Catch: Handles exceptions if it occurs

Finally: Executes always irrespective of exception occurs or not usually for resource cleanup

Part - B.

1. Concept of class, object, constructor and method with example program

Class:

- * A class is a blueprint or a template
- * A collection of objects

Syntax: class className

```
{  
    dataType variableName;  
    returnType methodName()  
}
```

eg:

```
import java.io.*;  
class A  
{  
    int a; dataType variable name;  
    void method name()  
}
```

Object

- Object are real world entity

An object is an Instance of a class

An object consists of

* State

* Behaviour

Syntax:

```
className objectName = new  
className();
```

Example program:

```
import java.io.*;  
class A  
{  
    int a;  
    datatype variableName;  
    void methodName()  
}  
  
class demo  
{  
    public static main(String args[])  
    {  
        A a = new A();  
    }  
}
```

```
2 methodname();
```

Constructor :

Constructor is a special type of method that can be used to initialize an object.

Syntax:

```
[access specifier] class name ([arguments])
```

```
{  
    statements;  
}
```

Types of constructor

No argument constructor

Parameterized constructor

Default constructor

No argument Constructors:

Do not accept any parameters

class demo

```
{
```

```
int a;
```

```
demo() {
```

```
}
```



```

a = 100 ;
}
void display ()
{
System.out.println(a);
}
}
public class Sample
{
public static void main (String args [])
{
demo d = new demo();
d.display();
}
}

```

Output :

100

Parameterized constructor

Constructor that accepts one or more parameters

class demo

```

{
int a;
demo(int n)
{

```

a = n;

```

void display()
{
    System.out.println("Number is" + a);
}
}
public class sample
{
    public static void main (String args[])
    {
        demo d = new demo();
        d.display();
    }
}

```

Output:

100

Constructor overloading:

Overloaded constructors are different in their type of parameters or number of parameters

class demo

{

int a;

demo()

{

a = 100;

}

demo(int b)

```

{
a = n;
}
void display ()
{
System.out.println ("Number is "+ a);
}
}

public class Sample
{
public static void main (String args [])
{
demo t1 = new demo();
demo t2 = new demo(200)
t1.display();
t2.display();
}
}

```

Output:

Number is 10

Number is 200

Methods:

Functions defined inside the class definition are called as methods

Syntax of methods

```
[access-specifier] [non access specifier] <type>  
-> function name (parameters list)  
{  
    body of the function/method;  
}
```

Methods without parameters:

```
class Sum  
{  
    int a, b, c;  
    void add ()  
    {  
        a = 5;  
        b = 50;  
        c = a + b;  
        System.out.println ("The sum is: " + c);  
    }  
}
```


class Sample

```
public static void main (String [] args)
```

```
Sum s = new Sum ();
```

```
add ();
```

Methods with parameters:

class Sum

```
{  
    int a, b, c;
```

```
    void add (int x, int y)
```

```
{  
        a = x;
```

```
        b = y;
```

```
        c = a + b;
```

```
        System.out.println ("The sum is: " + c);
```

```
    }
```

```
}
```

class Sample

```
{  
    public static void main (String [] args)
```

```
    Sum s = new Sum ();
```

S. add (100, 200);

4

4

2. Different types of Inheritance :

The process of deriving a new from an old program is called Inheritance.

Old class of java is called as super class or parent class and new class of java is called as subclass/child class.

Single Inheritance (only one superclass)

Hierarchical Inheritance (one superclass, multiple subclasses)

Multilevel Inheritance (derived from a derived class)

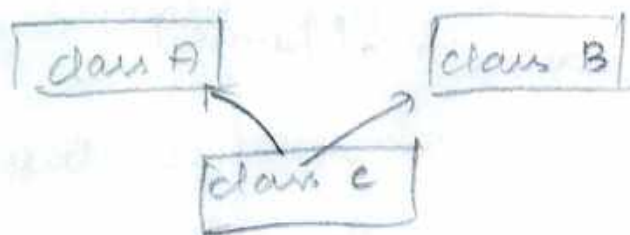
Single



Hierarchical Inheritance



Multilevel Inheritance



Single Inheritance

Child Inherits from Parent

// Single Inheritance

class Parent

{
 void displayParent()

{

System.out.println("This is parent class");

}

class Child extends Parent

{
 void displayChild()

```
System.out.println("This is Grand Child class (Single Inheritance)");
```

```
public class Inheritance Demo
```

```
{  
    public static void main (String[] args)
```

```
{  
    Child c = new Child();
```

```
    c.displayParent();
```

```
    c.displayChild();
```

```
}
```

Hierarchical Inheritance

Class AnotherChild extends Parent

```
{  
    void displayAnotherChild()
```

```
{  
    System.out.println("This is AnotherChild class  
    (Hierarchical Inheritance)");
```

```

AnotherChild gc = new AnotherChild ();
gc.displayParent ();
gc.displayAnotherChild ();
}
}

```

~~multicultural~~ Inheritance:

Grand child / In Multiple classes
AnotherChild inherits from one parent (Parent).

Multicultural Inheritance:

Grand child inherits from child
which inherits from Parent

```

Class Grand AnotherChild extends Child
{
    void displayGrandChild ();
}
System.out.println ("This is Grand child class  
(Multicultural Inheritance)");
}
}

```

```

GrandChild gc = new GrandChild ();
gc.displayParent ();
gc.displayChild ();
}
}

```


Q.3. Difference between checked and unchecked exceptions.

Checked exceptions and unchecked exceptions

Checked exceptions

Checked exceptions are checked at compile time.

It means if a method is throwing a checked exception then it should handle the exception using try-catch block or should declare the exception using throws keyword otherwise the program will give compilation error.

Example:

try-catch block:

```
try {  
    // code that throws exception  
} catch (Exception e) {  
    // handle exception  
}
```

or

throws declaration:

```
public void readData() throws IOException {  
    // code that throws exception  
}
```

or

or

or

```

try {
    Reader fr = new FileReader("missing.txt");
}
catch (FileNotFoundException e)
{
    System.out.println("Caught: " + e);
}

try {
    Class.forName("MissingClass");
}
catch (ClassNotFoundException e)
{
    System.out.println("Caught: " + e);
}

System.out.println("Program completed.");

```

Unchecked exceptions:

Unchecked Exceptions are not checked at compile time

It means if your program is throwing an unchecked exception and even if you didn't handle / declare that exception the program won't give a compilation error.

2.11 unchecked exceptions are direct
sub classes of Runtime exception class

Class Unchecked Exceptions All

```
↓  
public static void main (String [] args)
```

```
{  
    1. Arithmetic Exception.
```

```
    try  
    {  
        int result = 10/0;  
    }
```

```
    catch (ArithmeticException e)
```

```
{  
    System.out.println ("caught: " + e);  
}
```

```
2. Null Pointer Exception.
```

```
try  
{  
    String str = null;
```

```
    System.out.println (str.toLowerCase());  
}
```

```
catch (NullPointerException e)
```

```
{  
}
```

```
System.out.println("caught:" + e);
```

```
}
```

3. Array Index Out of Bounds Exception.

```
try
```

```
{
```

```
int[] arr = {1, 2, 3, 4};
```

```
System.out.println(arr[5]);
```

```
}
```

```
catch (ArrayIndexOutOfBoundsException e)
```

```
{
```

```
System.out.println("caught:" + e);
```

```
}
```

4. String Index Out of Bounds Exception

```
try
```

```
{
```

```
String s = "Hi";
```

```
System.out.println(s.charAt(5));
```

```
}
```

```
catch (StringIndexOutOfBoundsException e)
```

```
{
```

```
System.out.println("caught:" + e);
```

```
}
```

5.1 Number Found Exception

try

{

int num = Integer.parseInt("abc");

}

catch (NumberFormatException)

{

System.out.println("Caught: " + e);

}

S.A.ENGINEERING COLLEGE, CHENNAI - 77
(An Autonomous Institution, Affiliated to Anna University)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
REMEDIAL CLASS TIME TABLE

DATE:5.3.2024

ACADEMIC YEAR:2023-2024 EVEN

Note: Failures based on **Internal Assessment - 1** has to attend the remedial class and attendance for the same has to be recorded in log book under NSLB.

Sl. No	Date	Year/ Sem	Subject Code/Name	Name of the Staff		
				A Sec	B Sec	C Sec
1	5.3.24 Dr.D.Hemanand	III YEAR	CS1601A- Internet Programming	Ms.V.Saraswathi	Mr.S.Muthukumara samy	Ms.C.Lakshmi priya
2	6.3.24 Dr.D.Chitradevi	III YEAR	CS1602A- Compiler Design	Nil	nil	Ms.J.Sangeetha
	7.3.24 Mrs.Saraswathy V	III YEAR	CS1602A- Compiler Design	Ms.T.Thilagam	Dr.M.Nalini	nil
3	8.3.24 Dr.M.Nalini	III YEAR	CS1603A- Artificial Intelligence	Dr.R.Geetha	Dr.M.Anuradha	Dr.N.Thilagavathi


Dr. N. Thilagavathi

S.A.ENGINEERING COLLEGE, CHENNAI - 77
(An Autonomous Institution, Affiliated to Anna University)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
REMEDIAL CLASS TIME TABLE

ACADEMIC YEAR:2023-2024 EVEN

DATE:5.3.2024

Note: Failures based on **Internal Assessment - 1** has to attend the remedial class and attendance for the same has to be recorded in log book under NSLB.

Sl. No	Date	Year/ Sem	Subject Code/Name	Name of the Staff		
				A Sec	B Sec	C Sec
1	5.3.24 Dr.D.Hemanand	III YEAR	CS1601A- Internet Programming	Ms. V. Saraswathi	Mr.S.Muthukumarasamy	Ms.C.Lakshmi priya
2	6.3.24 Dr.D.Chitradevi	III YEAR	CS1602A- Compiler Design	Nil	nil	Ms.J.Sangeetha
	7.3.24 Mrs.Saraswathy V	III YEAR	CS1602A- Compiler Design	Ms. T. Thilagam	Dr.M.Nalini	nil
3	8.3.24 Dr.M.Nalini	III YEAR	CS1603A- Artificial Intelligence	Dr.R.Geetha	Dr.M.Anuradha	Dr.N.Thilagavathi
4.	11.3.24 Mr.S.Muthukumarasamy	III YEAR	CS1604A- Cryptography and Network Security	Dr.D.Chitradevi	Dr.D.Hemanand	Dr.A.Antonydoss
5	18.3.24 Mr.Kaliraj	III YEAR	CS1608A -Cyber Security	Mr.C.Balakrishnan Ms.M.Kavitha Ms.J.Rajalakshmi		
		III YEAR	CS1609A- Software Testing	Ms. V. Kavitha		


HOD

S.A.ENGINEERING COLLEGE, CHENNAI – 77
(An Autonomous Institution, Affiliated to Anna University)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
REMEDIAL CLASS TIME TABLE

ACADEMIC YEAR:2023-2024 EVEN

DATE:18.3.2024

Note: Failures based on Internal Assessment – 1 has to attend the remedial class and attendance for the same has to be recorded in log book under NSLB.

Sl. No	Date	Year/ Sem	Subject Code/Name	Name of the Staff		
				A Sec	B Sec	C Sec
1	20.3.24 Mrs.M.Kavitha	II YEAR	HV1401A-Universal Human Values	Mr.C.Balakrishnan	Dr.Muthumariakshmi	Ms.S.Gayathri
2	21.3.24 Ms.Sudha L	II YEAR	CS 1401A- Computer Architecture	Ms.S.N.Ananthi	Mr.V.Kaliraj	Ms.K.B.Aruna
		III YEAR	CS1601A- Internet Programming	Ms.V.Saraswathi	Mr.S.Muthukumarasa my	Ms.C.Lakshmi priya
		II YEAR	CS 1403A-Object Oriented Analysis and design	Mr.M.Balasubramanian	Ms.Belsinia Jebamalar	Ms.S.Arumanai Shiney
3	22.3.24 Ms.Sureka V	III YEAR	CS1602A- Compiler Design	Ms.T.Thilagam	Dr.M.Nalini	Ms.J.Sangeetha
4	25.3.24 Ms.Aruna.K.B	II YEAR	MA 1403A-Probability and Statistics	Dr.Loganathan	Ms.Selvi	Ms.Shoba
		III YEAR	CS1603A- Artificial Intelligence	Dr.R.Geetha	Dr.M.Anuradha	Dr.N.Thilagavathi
		II YEAR	IT 1401 A-Database Management Systems	Mr.A.Mani	Mr.M.Balasubramanian	Ms.K.Jeyanthi
5	26.3.24 Ms.S.Arumanai Shiney	III YEAR	CS1604A- Cryptography and Network Security	Dr.D.Chitradevi	Dr.D.Hemanand	Dr.A.Ananyadoss
6.	8.04.24 Mr.Prasanth	II YEAR	CS 1402A-Design And Analysis Of Algorithms	Dr.S.Sivakumar	Ms.L.Sudha	Ms.V.Sureka


HOD

S.A.ENGINEERING COLLEGE, CHENNAI - 77
(An Autonomous Institution, Affiliated to Anna University)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UPDATED REMEDIAL CLASS TIME TABLE

DATE:25.3.2024

ACADEMIC YEAR:2023-2024 EVEN

Note: Failures based on Internal Assessment - 1 has to attend the remedial class and attendance for the same has to be recorded in log book under NSLB.

Sl. No	Date	Year/ Sem	Subject Code/Name	Name of the Staff		
				A Sec	B Sec	C Sec
1	20.3.24 Mrs.M.Kavitha	II YEAR	HV1401A-Universal Human Values	Mr.C.Balakrishnan	Dr.Muthumariakshmi	Ms.S.Gayathri
2	21.3.24 Ms.Sudha L	II YEAR	CS 1401A- Computer Architecture	Ms.S.N.Ananthi	Mr.V.Kaliraj	Ms.K.B.Aruna
		III YEAR	CS1601A- Internet Programming	Ms.V.Saraswathi	Mr.S.Muthukumarasamy	Ms.C.Lakshmiyaya
3	22.3.24 Ms.Sureka V	II YEAR	CS 1403A-Object Oriented Analysis and design	Mr.M.Balasubramanian		Ms.S.Arumanai Shiney
		III YEAR	CS1602A- Compiler Design	Ms.T.Thilagam	Dr.M.Nalini	Ms.J.Sangeetha
4	8.4.24 Ms.Aruna.K.B	II YEAR	MA 1403A-Probability and Statistics	Dr.Loganathan	Ms.Selvi	Ms.Shoba
5	10.4.24 Ms.S.Arumanai Shiney	II YEAR	IT 1401 A-Database Management Systems			Ms.K.Jeyanthi
			CS 1403A-Object Oriented Analysis and design		Ms.Belsia Jebamalar	
6.	12.04.24 Ms.Sreevani	II YEAR	CS 1402A-Design And Analysis Of Algorithms			Ms.V.Sureka
7.	15.4.24 Mr.Prasanth	II YEAR	IT 1401 A-Database Management Systems	Mr.A.Mani	Mr.M.Balasubramanian	
			CS 1402A-Design And Analysis Of Algorithms	Dr.A.Antonydoss	Ms.L.Sudha	

MOD

S.A.ENGINEERING COLLEGE, CHENNAI - 77.
 (An Autonomous Institution, Affiliated to Anna University)
REMEDIAL CLASS ATTENDANCE

DEPT: CSE

DATE: 8.3.24

SL.NO	YEAR / SECTION	STAFF NAME	SUBJECT NAME	TOTAL NO OF STUDENTS	NO OF STUDENTS PRESENT	NO. OF STUDENTS ABSENT		SIGNATURE OF THE SUBJECT INCHARGE
						FULL DAY ABSENT	FOR REMEDIAL CLASS ALONE ABSENT	
1	II/A							
2	II/B							
3	II/C							
4	III/A	Dr. R. Geetha	Artificial Intelligence	16	9	03 (19, 34, 75)	4 (8, 40, 11, 171) (1908 p.m.)	<i>[Signature]</i>
5	III/B	Dr. M. Anusree	Artificial Intelligence	16	10	19, 34, 75 103	—	<i>[Signature]</i>
6	III/C	Dr. N. Thirugavathi	Artificial Intelligence	15	15	—	—	<i>[Signature]</i>
7	IV/A							
8	IV/B							
9	IV/C							

STAFF INCHARGE/HOD

PRINCIPAL

S.A. ENGINEERING COLLEGE, CHENNAI - 77.
(An Autonomous Institution, Affiliated to Anna University)
REMEDIAL CLASS ATTENDANCE

DEPT: CSE

DATE: 21/03/24

SI NO	YEAR / SECTION	STAFF NAME	SUBJECT NAME	TOTAL NO OF STUDENTS	NO OF STUDENTS PRESENT	NO. OF STUDENTS ABSENT		SIGNATURE OF THE SUBJECT INCHARGE
						FULL DAY ABSENT	FOR REMEDIAL CLASS ABSENT	
1	I/A	S.N. Ananthi	CA	15	11	04 (12, 28, 47, 49) 05	—	Posa
2	I/B	Dr. M. Anusupama	CA	11	6	15, 84, 92, 104, 307, 32	—	M. Anusupama
3	I/C	S. Anand Shriv	DOAB	10	8	17, 143	—	T. Anand
4	II/A	V. Sarabathi	IP	19	18	—	118 (lost prior permission) 93 (lost prior permission)	V. Sarabathi
5	II/B	S. Manikam	IP	11	10	—	—	S. Manikam
6	II/C	C. Lakshmi Priya	IP	21	21	—	—	C. Lakshmi Priya
7	IV/A							
8	IV/B							
9	IV/C							

21/03/24
STAFF INCHARGE

PRINCIPAL

BAND WIDTH ANALYSIS / REVIEW ACTION PLAN REPORT

MM F05B

Year / Sem: III / I

NSLB - REMEDIAL CLASS ASSESSMENT REPORT (UT 1)

CRP F07

Course / Year: BE / III

Subject: A2

Date of Submission: 11/2/24

Dept: CSE
 Subject: Artificial Intelligence Subject Code: CA16057

CLASSES	IA 1	IA 2	IA 3	UNIVERSITY EXAM
Highest Mark	90	90	92	0
Lowest Mark	02	30	06	C
Subject Average	59.68	62.22	60.52	A
Pass %	71.78	82.53	84.13	100
No. of Students Scored (100)	-	-	-	-
No. of Students Scored (91-99)	-	-	02	01
No. of Students Scored (81-90)	08	08	06	07
No. of Students Scored (71-80)	12	13	15	31
No. of Students Scored (61-70)	15	11	11	16
No. of Students Scored (50-60)	14	20	19	08
No. of Students Scored < 50	11	10	10	-
Absentees	03	01	-	-
Total Class Strength	63	63	63	63
Signature of Staff Incharge				
Signature of HOD				

Review Commentary :

Roll No. / Reg. No.	Name	UT 1 Marks	Attendance
---------------------	------	------------	------------

13	Balaji	44	
21	Bharathi	48	
128	Sanjay Kumar	46	
153	Sandeep Kumar	49	
161	Syed Abbas	46	
162	Pranil Verma	46	

No. Present	6
No. Absent	
Faculty Initials	

TOPICS DISCUSSED / ACTION PLAN

Assig ment given on regularity

S.A.ENGINEERING COLLEGE, CHENNAI - 77
(An Autonomous Institution, Affiliated to Anna University)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
RESULT Analysis for subject less than 90%

Academic Year: 2023-2024 (EVEN)

Year/Sem: III / VI

Section: 'A'

Branch: CSE

Subject Code: CS1603 A

Subject Name: Artificial Intelligence

Faculty Handled: Dr. R. Gopalak

S.NO	REG NO	NAME OF STUDENT	UNIT TEST-1	IA-1	UNIT TEST-2	IA-2	IA-3	AU RESULT BEFORE RV	AU RESULT AFTER RV
1.	111921CS01 1013	D. Balaji	44	46		50	29	A	
2	1021	C.R. Bharath	48	50		58	54	A	
3	1128	G. Sanjay Kumar	46	70		58	64	A	
4	1153	B. Sugumarwar	48	68		62	81	A	
5	1161	S.H. Syed Akbar	46	42		50	59	B+	
6	1162	T. Tamilvaran	44	54		32	41	C	
7	08	Alistar Francis	56	48		50	20	B	
8	40	Gajalakshmi	82	02		18	42	A	
9	92	Marayanan	68	44		42	50	B+	
10	62	Prabhu H	62	36		32	35	B	
11	105	Pragadish N	54	34		50	19	B	
12	111	Priyadarshini H	88	48		18		A	
13	151	Sriramamni Bagar	54	34		50	06	B	

S.NO	REG NO	NAME OF STUDENT	UNIT TEST-1	IA-1	UNIT TEST-2	IA-2	IA-3	AU RESULT BEFORE RV	AU RESULT AFTER RV
14	165	Tharun VS	70	83		36	54	B+	
15	171	Varsha	78	83		64	95	A	
16	175	Vignesh PK	56	38		34	50	B+	
17	301	Daran Harshad	80	48		30	50	B	
18	303	Kannan	58	28		42	51	B	
19	127	Satthi Nishanth	70	56		42	109	B+	
20	163	Tarunika SS	86	64		42	54	A	
21	177	Vikraman T	74	74		44	69	A	
22	29	Chikha	88	78		12	33	A	
23	50	Jay Shwanth	84	62		56	86	B+	
24	68	Kirthi Varan	70	64		84	157	B+	

TEST	FACULTY REMARKS	SIGNATURE	HODs REMARKS	SIGNATURE
UNIT -1	6 Students failed Pass % = 90.48 AB-M1 (Failed student scored above 44)		Good	
IA-1	11 Students failed, AB-03 Pass % = 77.78 Failed student attempted problem in Unit 2 and solved incorrectly		Give more programme	
UNIT -2				
IA-2	Pass % = 82.53 10 Students failed, AB-01. All failures have scored above 32 marks, except explanation was not clear		Good	
IA-3	Pass % = 84.13 10 Students failed, Due to placement drive, 10 Students did not present		Good	
AU	Pass % = 100 All student cleared my subject		Good	

Unit 2

1. Single inheritance:

```

class A
{
    int a;
    void methodname()
}

class B extends A
{
    int b;
    void methodname2()
}

class demo
{
    public static main(String args[])
    {
        B b=new B();
        b.methodname();
        b.methodname2();
    }
}
    
```

2. Multilevel inheritance:

```

class A
{
    int a;
    void methodname()
}
    
```

```

void methodname3()
{
}
    
```

```

class demo
    
```

```

{
    public static main(String args[])
    {
    
```

```

        C c=new C();
    
```

```

        c.methodname();
    
```

```

        c.methodname2();
    
```

```

        c.methodname3();
    }
}
    
```

3. Hierarchical inheritance:

```

class A
    
```

```

{
    
```

```

    int a;
    
```

```

    void methodname()
    {
    
```

```

    }
    
```

```

class B extends A
    
```

```

{
    
```

```

    int b;
    
```

```

    void methodname2()
    {
    
```

```

    }
    
```

```

class C extends A
    
```

```

{
    
```

```

    int b;
    
```

```

    void methodname3()
    {
    
```

```

    }
    
```

```

class B extends A
    
```

```

{
    
```

```

    int b;
    
```

```

    void methodname3()
    {
    
```

```

        b.methodname();
    
```

```

        b.methodname3();
    
```

```

        C c=new C();
    
```

```

        c.methodname();
    
```

```

        c.methodname3();
    }
}
    
```

4. Abstract:

```

abstract class A
    
```

```

{
    
```

```

    int a;
    
```

```

    abstract void methodname();
    
```

```

    class B extends A
    {
    
```

```

        void methodname()
    {
    
```

```

    }
    
```

```

class demo
    
```

```

{
    
```

```

    public static main(String args[])
    {
    
```

```

        B b=new B();
    
```

```

        b.methodname();
    }
}
    
```

```

class B extends A
    
```

```

{
    
```

```

    void methodname()
    {
    
```

```

    }
    
```

```

class demo
    
```

```

{
    
```

```

    public static main(String args[])
    {
    
```

```

        B b=new B();
    
```

```

        b.methodname();
    }
}
    
```

```

class B extends A
    
```

```

{
    
```

```

    void methodname()
    {
    
```

```

    }
    
```

```

class demo
    
```

```

{
    
```

```

class A
{
    public static void main(String args[])
    {
        int d, a;
        try
        {
            ...
        }
        catch (Exception e)
        {
        }
    }
}

```

2. Multiple exception:

```

class A
{
    public static void main(String args[])
    {
        try
        {
            ...
        }
        catch(ArithmeticException e)
        {
            ...
        }
        catch(ArrayIndexOutOfBoundsException e)
        {
            ...
        }
    }
}

```

3. User defined Exception:

```

import java.util.*;
class A extends Exception
{
    A()
    {
    }
    public String toString()
    {
        return msg;
    }
}

```

```

class Demo
{
    public static void main(String args[])
    {
        try
        {
            ...
        }
        throw new A();
    }
    catch (A e)
    {
    }
}

```

4. Input:

```

Class A
{
    public static void main(String args[]) throws
    IOException
    {
        int a;
        BufferedReader br = new BufferedReader(new
        InputStreamReader(System.in));
        a = Integer.parseInt(br.readLine());
    }
}

```

5. File Input:

```

import java.io.*;
class A
{
    public static void main(String args[])
    {
        try
        {
            FileInputStream fin = new
            FileInputStream("k:/file.txt");
            while((c = fin.read()) != -1)
            {
            }
        }
    }
}

```

```

        fin.close();
    }
    catch(IOException e)
    {
    }
}

```

1. Thread:

Class A implements Runnable

Unit 4

```
public void run()
```

```
public static void main (String [] args)
```

```
A t = new A();
```

```
Thread t1 = new Thread(r, "Thread-A");
```

```
Thread t2 = new Thread(r, "Thread-B");
```

```
t1.start();
```

```
t2.start();
```

2. Thread:

public class A extends Thread

```
public void run()
```

```
public static void main(String args[])
```

```
A t1 = new A();
```

```
A t2 = new A();
```

```
t1.start();
```

```
t2.start();
```

3. Synchronized:

class A implements Runnable

```
public void run()
```

```
public static void main (String [] args)
```

```
A nr = new A();
```

```
Thread one = new Thread(nr, "Thread-A");
```

```
Thread two = new Thread(nr, "Thread-B");
```

```
one.start();
```

```
two.start();
```

4. Inter-thread communication:

```
class A {
```

```
public static void main(String [] args)
```

```
B b = new B();
```

```
b.start();
```

```
synchronized(b)
```

```
{
```

```
try
```

```
{
```

```
b.wait();
```

```
catch (InterruptedException e)
```

```
{
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
Box(T t)
```

```
{
```

```
a = t;
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

6. Generic method:

```
class Box<T>
```

```
{
```

```
private T a;
```

```
void assign(T t)
```

```
{
```

```
a = t;
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```

```
}
```


1.Thread:
Class A implements Runnable

Unit 4

```
{
    public void run()
    {
    }

    public static void main (String [] args)
    {
    }
}
```

```
A r = new A();
Thread t1 = new Thread(r, "Thread-A");
Thread t2 = new Thread(r, "Thread-B");
```

```
t1.start();
t2.start();
}
```

2.Thread:

```
public class A extends Thread
{
    public void run()
    {
    }
}
```

```
public static void main(String args[])
{
}
```

```
A t1 = new A();
A t2 = new A();
```

```
t1.start();
t2.start();
}
```

3.Synchronized:

```
class A implements Runnable
{
    public void run()
    {
    }
}
```

```
public static void main (String [] args)
{
}
```

```
A nr = new A();
```

```
Thread one = new Thread(nr, "Thread-A");
```

```
Thread two = new Thread(nr, "Thread-B");
```

```
one.start();
```

```
two.start();
}
```

4.Inter-thread communication:

```
class A {
```

```
    public static void main(String [] args)
    {
    }
```

```
B b = new B();
```

```
b.start();
```

```
synchronized(b)
{
}
```

```
try
```

```
{
}
```

```
b.wait();
}
```

```
catch (InterruptedException e)
{
}
```

```
}
}
```

```
}
}
```

```
}
}
```

```
}
}
```

```
class B extends Thread
{
}
```

```
    public void run() {
        synchronized(this) {
            notify();
        }
    }
}
```

```
    notify();
}
```

```
    }
}
```

Box(T t)

```
{
    a = t;
}
```

```
}
}
```

```
class Main
{
}
```

```
    public static void main(String[] args)
    {
    }
```

```
        Box<Integer> b1 = new Box<Integer>(10);
```

```
        Box<String> b2 = new Box<String>("Hello");
```

```
        b1.print();
```

```
        b2.print();
    }
}
```

6.Generic method:

```
class Box<T>
```

```
{
}
```

```
    private T a;
```

```
    void assign(T t)
    {
    }
```

```
        a = t;
    }
}
```

```
class Main
{
}
```

```
    public static void main(String[] args)
    {
    }
```

```
        Box b1 = new Box();
```

```
        Box b2 = new Box();
```

```
        b1.<Integer>assign(10);
```

```
        b2.<String>assign("Hello");
```

```
        b1.print();
```

```
        b2.print();
    }
}
```



PARENT TEACHERS MEETING

FOR THIRD YEAR CSE – A, B, C STUDENTS



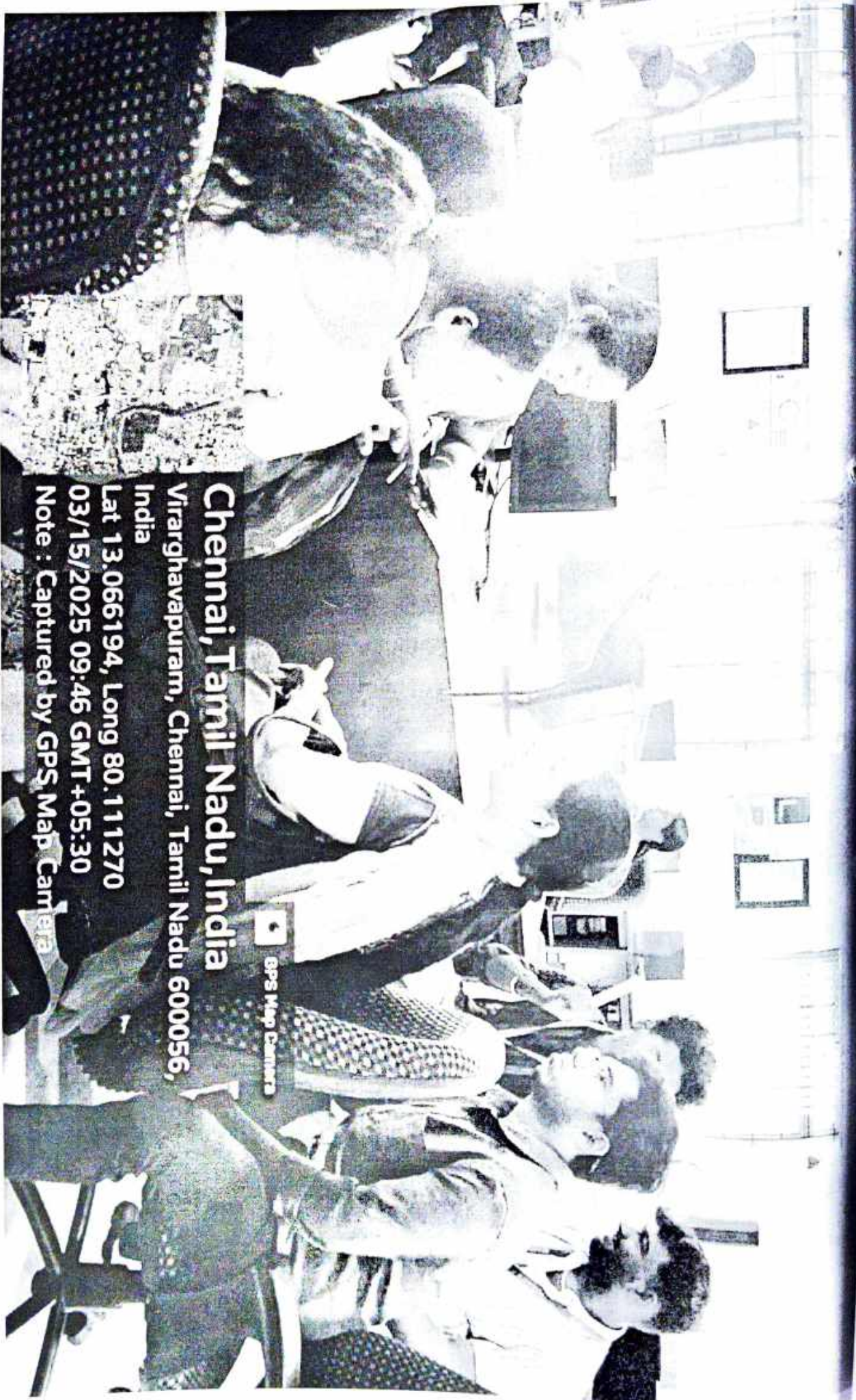
R&D LAB



8:30 AM – 4:05 PM



15/03/2025



Chennai, Tamil Nadu, India

Virarghavapuram, Chennai, Tamil Nadu 600056,

India

Lat 13.066194, Long 80.111270

03/15/2025 09:46 GMT+05:30

Note : Captured by GPS Map Camera



S.A. ENGINEERING COLLEGE, CHENNAI-77.

CRP 409

(An Autonomous Institution, Affiliated to Anna University)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Mentoring Efficacy

Academic Year/ Sem : 2024-2025/ VIII

Name of the Mentor: Ms.J Sangeetha

Class: IV B

S.N	Name of the Student with Register Number	Counselling I (Purpose of Counselling)	Counselling II (Purpose of Counselling along with Follow-up of Counselling I)	Counselling III (Purpose of Counselling Along with Follow-up of Counselling II)	Overall Improvement through Counselling
0					
1	KISHORE S 111921CS01071	General Counselling. Motivated for placements. Discussed about University Results	Encouraged him to concentrate on academics as well as placements.	Advised him to focus on studies & score high CGPA	Got placed in US Based Company
2	P JAYALAKSHMI 111921CS01057	General Counselling. Motivated for placements. Discussed University Results.	Encouraged her to concentrate on academics as well as placements.	Advised her to focus on studies & score high CGPA	Got placed in Domain Company.
3	111921CS01048 P HARINI	General Counselling. Motivated for placements and Attendance. Discussed about University Results	Encouraged her to concentrate on academics as well as placements.	Motivated her to do internship and projects.	Got placed in SUTHERLAND
4	111921CS01047 A HARINI	General Counselling. Motivated for placements. Discussed University Results.	Encouraged her to concentrate on academics as well as placements.	Advised her to focus on studies & score high CGPA	Got placed in CROYEZ

5	111921CS01090	General Counselling. Motivated for placements. Discussed about University Results	Encouraged her to concentrate on academics as well as placements.	Advised her to focus on studies & score high CGPA	Motivated to do hard work, and do online courses. Avoid taking leave
6	111921CS01075 LIKITH P	General Counselling. Motivated for placements and Attendance. Discussed about University Results	Encouraged him to concentrate on academics as well as placements.	Motivated him to do internship and projects solve the skill rack problem on time.	Improved performance in skill rack platform and scored good marks in assessments.
7	111921CS01064 G KARPAGALAKSHMI	General Counselling. Motivated for placements. Discussed about University Results	Discussed about IA-1 Performance. Advised to be regular to college.	Advised her to search for internship opportunities and to do projects	Need to concentrate even more on placements
8	111921CS01056 S A JAYAKUMAR	General Counselling. Motivated for placements and Attendance. Discussed about University Results	Discussed about IA-1 Performance. Advised him to attend classes on time and to be regular to college	Advised him to improve in academics	Should concentrate more on academics
9	111921CS01073 L LALITH LARESH	Motivated to do hard work, attend placement training and avoid taking leave.	Motivated to do hard work, attend class regularly and score good mark in IA2. avoid taking leave.	Motivated to do hard work, and do online courses and solve the skill rack problem on time.	Improved performance in skill rack platform and scored good marks in assessments.
10	111921CS01064 M KANISHKA	General Counselling. Motivated for placements. Discussed University Results.	Encouraged her to concentrate on academics as well as placements.	Advised her to focus on studies & score high CGPA	Got placed in Moveate

Counseling Criteria for Enhancing Student's Performance

1. Punctuality/Class Room Attentiveness
2. Attendance
3. University Exam Arrears
4. Internal Assessment Test Arrears
5. Dress Code, Cleanliness & Obedience
6. Motivation towards Higher Studies/Entrepreneur/Placement
7. Motivation towards Co-curricular activities
8. Motivation towards University Rank
9. Motivation towards Sports Activities

Signature of the Mentor

kraniumhealth



JASWANTH V

ASSISTANT SOFTWARE ENGINEER

EMP ID: KRA111

MOBILE: +91 9014681556

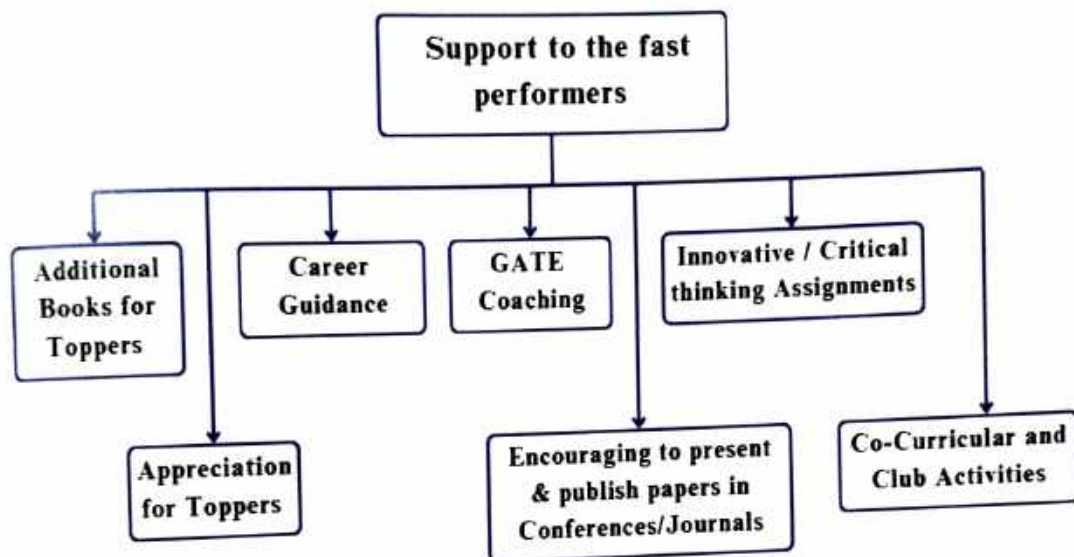


S.A ENGINEERING COLLEGE, CHENNAI - 77

(Autonomous - Institute Level Research Centre, Affiliated to Anna University)

Department of Computer Science and Engineering

SUPPORT TO THE FAST PERFORMERS





S.A. ENGINEERING COLLEGE, CHENNAI-77

(An Autonomous Institution, Affiliated to Anna University)

Accredited by NAAC 'A' Grade & ISO 9001:2015 Certified Institution



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



ORGANIZES

IEEE DAY 2024

**09
OCT
2024**



SPECIAL EVENTS

1. Hackathon
2. Mind Masters
3. UI Design
4. Connections
5. Debugging

REGISTER HERE



III YEAR CSE STUDENTS

044 26801999

www.saec.ac.in

saec@saec.ac.in



**ORACLE
Academy**

**CISCO Networking
Academy**



IEEE

Payment details

Payment confirmed via UPI

IEEE - Citicon order code 1000373750367571638229

14/02/25 12:09 pm

PIN 1000373750367571638229
E-Wallet UPI QRCode
Original amount in USD \$ 21.24
Total amount ₹ 1,910.11

Payment details

Payment confirmed via UPI

IEEE - Citicon order code 1000373747976344109061

14/02/25 11:51 pm

PIN 1000373747976244109061
E-Wallet UPI QRCode
Original amount in USD \$ 21.24
Total amount ₹ 1,910.11

1. PREDESHA

2. POORNIMA

Payment details

Payment confirmed via UPI

IEEE - Citicon order code 1000374096079282700293

15/02/25 12:54 pm

PIN 1000374096079282700293
E-Wallet UPI QRCode
Original amount in USD \$ 21.24
Total amount ₹ 1,904.59

Payment details

Payment confirmed via UPI

IEEE - Citicon order code 1000374093866722668309

15/02/25 12:44 pm

PIN 1000374093866722668309
E-Wallet UPI QRCode
Original amount in USD \$ 21.24
Total amount ₹ 1,904.59

3. ROHIT

4. INDIRANI

Payment details

Payment confirmed via UPI

IEEE - Citicon order code 1000373747125960488453

15/02/25 11:47 pm

PIN 1000373747125960488453
E-Wallet UPI QRCode
Original amount in USD \$ 21.24
Total amount ₹ 1,910.33

Payment details

Payment confirmed via UPI

IEEE - Citicon order code 1000374094888626971653

15/02/25 12:49 pm

PIN 1000374094888626971653
E-Wallet UPI QRCode
Original amount in USD \$ 21.24
Total amount ₹ 1,904.59

5. KOMESHWARI

6. SIVADINESH



ABIRAMI J CSE 2021-2025

Certificate of Completion for
AWS Academy Graduate - AWS Academy Cloud Foundations

Course hours completed
20 hours

Issued on
03/23/2024

Digital badge
<https://www.credly.com/go/Z2tkVEBd>

AWS global certification



Student completed certification course on "Java Full stack" in Wipro



ORGANIZES

VALUE ADDED COURSE

ON

CYBER FORENSICS & INCIDENT RESPONSE: TACTICAL BOOTCAMP

HOSTED BY



Dr. Sundresan Perumal
Associate Professor.

Faculty of Science and Technology.

University Sains Islam Malaysia

Bandar Baru Nilai

Nilai, Negeri Sembilan,

Malaysia



III YEAR CSE & II YEAR MCA

STUDENTS



25, 26 & 28

AUG

2025

Venue : MCA Seminar Hall

Timing : 9.00 AM To 4:05 PM



Elite

NPTEL Online Certification

(Funded by the MoE, Govt. of India)

This certificate is awarded to

GAUTAM KUMAR G

for successfully completing the course

Data Base Management System

with a consolidated score of **83** %

Online Assignments	22.5/25	Proctored Exam	60/75
--------------------	---------	----------------	-------

Total number of candidates certified in this course: **6225**

Jan-Mar 2024

(8 week course)



Indian Institute of Technology Kharagpur



Prof. Haimanti Banerji
Coordinator, NPTEL
IIT Kharagpur



Roll No

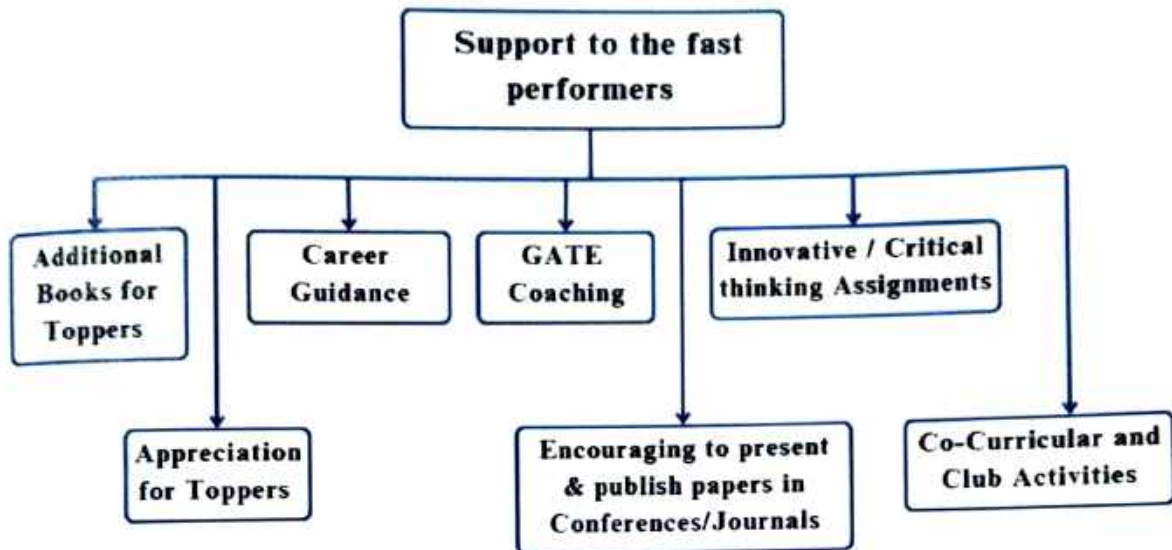
NPTEL24CS21S653403428

To verify the certificate



No. of credits recommended: 2 or 3

SUPPORT TO THE FAST PERFORMERS



change

S.A. ENGINEERING COLLEGE

Autonomous

Institute Level Research Center- Affiliated to Anna University, Chennai -600025

27th ANNUAL DAY

27th February 2025

Appreciation For Class Toppers - 2023-2024 Even

S.No.	DEPT	POSITION	COUNT	Amount in Rs.
1.	AIDS	I	02	Rs. 2,000/- + Certificate
		II	02	Rs. 2,000/- + Certificate
2.	AIML	I	01	Rs. 1,000/- + Certificate
		II	01	Rs. 1,000/- + Certificate
3.	Civil	I	02	Rs. 2,000/- + Certificate
		II	02	Rs. 2,000/- + Certificate
4.	CSBS	I	01	Rs. 1,000/- + Certificate
		II	01	Rs. 1,000/- + Certificate
5.	CSE	I	02	Rs. 2,000/- + Certificate
		II	02	Rs. 2,000/- + Certificate
6.	Cyber Security	I	NA	NA
		II	NA	NA
7.	ECE	I	02	Rs. 2,000/- + Certificate
		II	02	Rs. 2,000/- + Certificate
8.	EEE	I	03	Rs. 3,000/- + Certificate
		II	02	Rs. 2,000/- + Certificate
9.	H&S	I	12	Rs. 12,000/- + Certificate
		II	11	Rs. 11,000/- + Certificate
10.	IT	I	03	Rs. 3,000/- + Certificate
		II	02	Rs. 2,000/- + Certificate
11.	MECH	I	02	Rs. 2,000/- + Certificate
		II	02	Rs. 2,000/- + Certificate
12.	VLSI	I	NA	NA
		II	NA	NA
13.	MBA	I	01	Rs. 1,000/- + Certificate
		II	01	Rs. 1,000/- + Certificate
14.	MCA	I	01	Rs. 1,000/- + Certificate
		II	01	Rs. 1,000/- + Certificate
	TOTAL		61	61,000

I	32
II	29
Total	61

93000
61000
158000

Signature

S.A. ENGINEERING COLLEGE, CHENNAI-77
(AN AUTONOMOUS INSTITUTION, AFFILIATED TO ANNA UNIVERSITY)

CLASS WISE TOPPERS – (First 5 Places)

DATE: 30-03-2023

Degree : B.E
Branch : CSE
Academic Year : 2022 -2023
Semester : ODD

END SEMESTER EXAMINATIONS DEC 2022

S.NO.	YEAR/SECTION	REG. NO.	STUDENT NAME	GPA OBTAINED	POSITION
1	II A	111921CS01081	MADHUMITTA P	9.571	1
2		111921CS01028	CHARMILA A B	9.095	2
3		111921CS01131	SANJAY N	9.000	3
4		111921CS01149	SRIDEVI M	8.905	4
5		111921CS01043	GAYATHRI S	8.810	5
8	II B	111921CS01096	NIVEDITHA M	9.524	1
9		111921CS01097	PARVEEN BANU A	9.333	2
10		111921CS01003	ABIRAMI J	9.238	3
11		111921CS01086	MOHAMMED ANAS H	9.238	3
12		111921CS01088	MOHANKUMAR K	9.238	3
13		111921CS01112	PURUSOTHAMAN D	9.143	4
14		111921CS01170	VARSHA S	9.048	5
15	II C	111921CS01069	KIRUTHIGA S	9.286	1
16		111921CS01095	NITHYA SRI S	9.286	1
17		111921CS01129	SANJANA B L	9.143	2
18		111921CS01025	BLESSWIN BRIGHTSON G	9.095	3
19		111921CS01152	SRIVISHAG P	9.095	3
20		111921CS01174	VIGNESH P	9.000	4
21		111921CS01030	DEENA D	8.810	5
		111921CS01060	JONIE S	8.810	5
		111921CS01083	MAHALAKSHMI K	8.810	5
		111921CS01098	PAVITHRA V	8.810	5
22	III A	111920CS01014	BHUVANESHWARI N	9.348	1
23		111920CS01024	DIXITHA R	8.522	2
24		111920CS01037	HARSHAREEN TAJ S	8.435	3
25		111920CS01036	HARISH SRINIVAS S R	8.304	4
26		111920CS01009	ASHWIN KANTH H	8.261	5
27		111920CS01027	GANESH KUMAR K R	8.261	5
29	III B	111920CS01066	MADHAN T J	9.217	1
30		111920CS01072	MANOJ KUMAR M	8.783	2
31		111920CS01049	KAMALEE S	8.609	3
32		111920CS01052	KATHERINE KEZHIA VALAR	8.609	3
		111920CS01065	LOGESWARI P	8.609	3
		111920CS01078	NABISATHUL HAZEENA L A	8.478	4
		111920CS01086	NITHYA SHRI J	8.435	5
		111920CS01090	PRINCE MOSES D	8.435	5

34	III C	111920CS01109	SANTHOSH E	9.478	
35		111920CS01126	SURYA V	8.739	1
36		111920CS01138	VIGNESH S (12.12.2002)	8.652	2
37		111920CS01127	SWETHA B	8.348	3
38		111920CS01137	VIGNESH S (11.08.2002)	8.348	4
39		111920CS01096	RESHMA A	8.130	4
		111920CS01111	SATHISH KUMAR P	8.130	5
		111920CS01125	SURYA D	8.130	5
40	IV A	111919104059	KAVIYA R	9.318	5
41		111919104077	MONISHA S	9.318	1
42		111919104039	GAYATHRI PRRIYA D	9.182	1
43		111919104065	LEELARAMAN C	9.182	2
44		111919104131	SNEHA A	9.182	2
45		111919104135	SOWMIYA P	9.182	2
46		111919104093	PEMMASANI YASMITHA	9.045	3
47		111919104006	AKSHARA R	8.955	4
48		111919104052	JUSTUS KEVIN T	8.909	5
49		111919104053	KALAIVANI G	8.909	5
50		111919104096	POONTHAZHAI D	8.909	5
68	IV B	111919104142	SUBHASINI R	9.045	1
69		111919104120	SANTHOSHKUMAR B	8.818	2
70		111919104087	NILESH A	8.773	3
71		111919104103	PRIYADARSHINI D	8.773	3
72		111919104034	DIVYA C V S S	8.682	4
73		111919104165	YAMINI T	8.636	5
94	IV C	111919104146	SWETHA S	9.182	1
95		111919104009	ANITHA R	9.045	2
96		111919104044	HELEN ESTHER J N	9.045	2
97		111919104102	PRIYA S	9.045	2
98		111919104031	DINESH KUMAR D	8.955	3
99		111919104070	MARIYA ANTHONY	8.909	4
100		111919104086	NIKHITA M	8.909	4
101		111919104088	NIMISHAMBA K S	8.773	5
102		111919104123	SATHISHKUMAR K	8.773	5

HoD

PRINCIPAL



S.A. ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Approved by AICTE

Accredited by NAAC with 'A' Grade & ISO 9001:2015 Certified Institution

Poonamallee - Avadi Main Road, Thiruverkadu, Chennai - 600 077.

26th ANNUAL DAY

27th MARCH, 2024

CERTIFICATE OF EXCELLENCE

This is to certify that Mr. / Ms. NIVEDITHA M, III YEAR

of COMPUTER SCIENCE AND ENGINEERING department

has secured SECOND Place in V semester

during the Academic Year 2023-2024.

Dr. S. RAMACHANDRAN
Principal

Thiru. S. AMARNAATH
Correspondent





S.A. ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Approved by AICTE

Accredited by NAAC with 'A' Grade & ISO 9001:2015 Certified Institution
Ponnamallee - Avadi Main Road, Thiruvarkadu, Chennai - 600 077.

26th ANNUAL DAY

27th MARCH, 2024

CERTIFICATE OF EXCELLENCE

This is to certify that Mr. / Ms. PRAGADEESH D, II YEAR

of COMPUTER SCIENCE AND ENGINEERING Department

has secured SECOND Place in III semester

during the Academic Year 2023-2024.


Dr. S. RAMACHANDRAN
Principal


Thiru. S. AMARNAATH
Correspondent

www.saece.ac.in

Toppers certificate

S.A. ENGINEERING COLLEGE

Autonomous - Institute Level Research Centre- Affiliated to Anna University Chennai
Accredited by NAAC with 'A' Grade & ISO 9001:2015 Certified Institution



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GATE COACHING CLASS

DATE: 17-09-2025



Mrs.A.Shaini, Assistant Professor

Topic: Computer Networks-Application Protocol



Mrs.Lakshmi, Assistant Professor

Topic: Discrete mathematics - Set Theory



Mrs.S.Sasikala , Assistant Professor

Topic: Digital Logic and Boolean Expression-K Map



Mrs.R.Dhivyabharathi, Assistant Professor

Topic: DBMS-ER Model



Mrs.M.Revathi, Assistant Professor

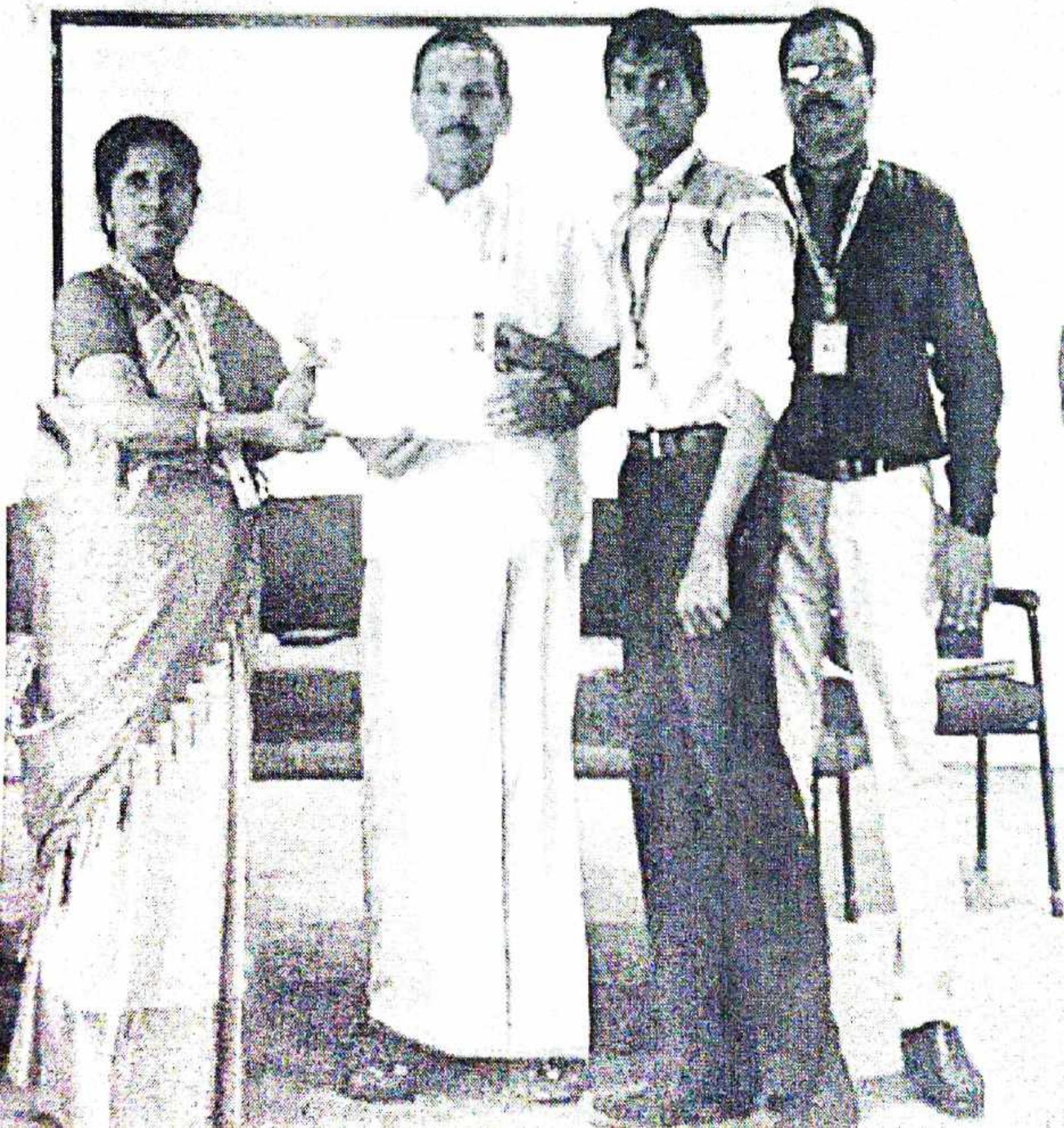
Topic: Operating System-Non Preemptive scheduling Algorithms

☎ - 044 26801999

🌐 - www.saec.ac.in

✉ - saec@saec.ac.in

U.S. GOVERNMENT
PROPERTY
NO LOAN OR REPRODUCTION
WITHOUT AUTHORITY



Advancing Cardiovascular Monitoring: Translation and Restoration of Photoplethysmogram using Super Operational Neural Networks in 1D-CycleGAN Framework

Publisher: IEEE Cite This PDF

R. Geetha ; M. Niveditha ; S. Revvina ; B. T. Sayvietha All Authors

6 Full Text Views

Abstract

Document Sections

I. Introduction

II. Literature Survey

III. Existing Methods

IV. Proposed System

V. Experimental Results

Show Full Outline

Authors

Figures

References

Keywords

Metrics

More Like This

Abstract:

Cardiovascular diseases endure as a predominant cause of death on a global scale, requiring precise monitoring methodologies. The current study confronts the issue of limited accuracy in estimating cardiovascular parameters through the introduction of a methodology that exploits Photoplethysmogram (PPG) signals. By employing a Translation Through Restoration GAN (TTR-GAN) framework in conjunction with Super Operational Neural Networks (Super-ONNs), the system adeptly translates and restores PPG signals from the wrist to the finger. Through extensive experimentation, a substantial 15% enhancement in accuracy is evident when compared to traditional methods such as logistic regression and support vector machines. Furthermore, the proposed system excels in terms of precision, recall, and the area under the receiver operating characteristic curve (AUC). This study presents a methodical approach to augmenting accuracy in the estimation of cardiovascular parameters, offering the potential for enhanced healthcare protocols, early complication identification, and improved patient care outcomes.

Published in: 2024 International Conference on Communication, Computing and Energy Efficient Technologies (I3CEET)

Date of Conference: 20-21 September 2024 **DOI:** 10.1109/I3CEET61722.2024.10993798

Date Added to IEEE Xplore: 13 May 2025 **Publisher:** IEEE

► ISBN Information: **Conference Location:** Gautam Buddha Nagar, India

Sign In to Continue Reading

Access to this document requires a subscription

IEEE offers both personal and institutional subscriptions. When you are an academic, a practitioner, or a student, IEEE offers a range of individual and institutional subscription options that meet your needs.

Authors

Figures

References



S.A. ENGINEERING COLLEGE
Autonomous Institution
Approved by M.A.E. Society & Government
Accredited by NMAC, South & Central
Registration No. 00011/M/2005

27th ANNUAL DAY CELEBRATION

27th ANNUAL DAY CELEBRATION





UDYAM REGISTRATION CERTIFICATE

UDYAM REGISTRATION NUMBER

UDYAM-TN-02-0322798

NAME OF ENTERPRISE

WEBZENITH SOLUTIONS

TYPE OF ENTERPRISE *

S.No.	Classification Year	Enterprise Type	Classification Date
1	2024-25	Micro	14/08/2024

MAJOR ACTIVITY

SERVICES

SOCIAL CATEGORY OF ENTREPRENEUR

GENERAL

NAME OF UNIT(S)

S.No.	Name of Unit(s)
1	WEBZENITH SOLUTIONS

OFFICIAL ADDRESS OF ENTERPRISE

Flat/Door/Block No.	Old No 17, New No 39	Name of Premises/ Building	1st Floor
Village/Town	Mylapore	Block	-
Road/Street/Lane	Muthukrishnan Street	City	Chennai
State	TAMIL NADU	District	CHENNAI, Pin 600004
Mobile	9514901537	Email:	webzenithsolutionschennai@gmail.com

DATE OF INCORPORATION / REGISTRATION OF ENTERPRISE

02/08/2024

DATE OF COMMENCEMENT OF PRODUCTION/BUSINESS

02/08/2024

S.No.	NIC 2 Digit	NIC 4 Digit	SIC 5 Digit	Activity
1	63 - Information service activities	6312 - Web portals	63122 - Operation of other websites that act as	Services

2	63 - Information service activities	6399 - Other information service activities n.e.c.	63999 - Other information service activities n.e.c.	Services
3	74 - Other professional, scientific and technical activities	7499 - Other professional, scientific and technical activities n.e.c.	74999 - Other professional, scientific and technical activities n.e.c.	Services
4	82 - Office administrative, office support and other business support activities	8299 - Other business support service activities n.e.c.	82999 - Other business support service activities n.e.c.	Services

14/08/2024

Disclaimer: This is computer generated statement, no signature required. Printed from: <http://www.mahatma.org.uk> & Date of printing- 14/08/2024

1. District Industries Centre:	CHENNAI (TAMIL NADU)
2. MSME-DFO:	CHENNAI (TAMIL NADU)



































BE A
CHAMPION
with the
Ministry of
MSME



Type of Enterprise		MICRO		Major Activity		Services	
Type of Organisation		Partnership		Name of Enterprise		WEBZENITH SOLUTIONS	
Owner Name		WEBZENITH SOLUTIONS		PAN		AAEFW9484G	
Do you have GSTIN		No		Mobile No.		9514901537	
Email Id		webzenithsolutionstech@gmail.com		Social Category		General	
Gender		Male		Specially Abled(DIVYANG)		No	
Date of Incorporation		02/08/2024		Date of Commencement of Production/Business		02/08/2024	

Bank Details	
Bank Name	IFS Code
STATE BANK OF INDIA	SBIN0002201
	Bank Account Number
	42973746437

Employment Details			
Male	Female	Other	Total
4	0	0	4

Investment in Plant and Machinery OR Equipment (in Rs.)

S.No.	Financial Year	Enterprise Type	Written Down Value (WDV)	Exclusion of cost of Pollution Control, Research &	Net Investment in Plant and Machinery OR Equipment[(A)-(B)]	Total Turnover (A)	Export Turnover (B)	Net Turnover [(A)-(B)]	Is ITR Filled?	ITR Type
-------	----------------	-----------------	--------------------------	--	---	--------------------	---------------------	------------------------	----------------	----------

600

District

Unit Name

Name of Premises/ Building

Flat/Door/Block No. Old No 17, New No 39

Nic 4 Digit

Nic 5 Digit

Date of Printing

Date of Udyam Registration

MSME-DFO

District Industries Centre

Are you interested in getting registered on Skill India Digital Portal

Are you interested in availing Free .IN Domain and a business email ID

Are you interested to get registered on NSIC B2B Portal

Are you interested to get registered on National Career Service(NCS) Portal

Are you interested to get registered on TReDS Portals(one or more)

Are you interested to get registered on Government e-Market (GeM) Portal

IEC Details

IEC Number

IEC Status

IEC Registration Date

IEC Modification Date

Inactive

₹. 100

Rs. 100

₹. 100

ONE

HUNDRED RUPEES

भारत INDIA

INDIA NON RESIDENTIAL

தமிழ்நாடு வணிகநாடு TAMILNADU

EF 4718977

1000 1000 1000 1000

1000 1000 1000 1000

1000 1000 1000 1000

NOW THIS DEED WITNESSES AS FOLLOWS:

1. The partnership shall be effective from 2nd day of AUGUST 2024.
2. That the business of the partnership shall be carried on under the name and style of "M/S WEBZENITH SOLUTIONS", (hereinafter referred as the "Firm") or such other name as may be mutually agreed upon from time to time.
3. That the Registered office of the Firm shall be at **OLD NO 17, NEW NO 39, MUTHU KRISHNAN STREET, MYLAPORE, CHENNAI 600004, TAMIL NADU** or any other place or places as the partners may decide from time to time.
4. The business of the partnership firm shall be to carry following activities:
 - a) To manufacture, sell, export, import all kinds of electric & electronic components capable of being used in Electrical & mechanical and electronic Industries including Computers telecommunications to carry out software research and development, to design and develop system software, application software and any other software in India and abroad to start integrated services Digital Local Network (ISDIAN) dial for data Centers technology parks, wide area network Internet, user net, cyber cafe services in India and abroad.

एक सौ रुपये

Rs. 100

भारत INDIA

नमिलनाडु TAMILNADU

Y. C. Gangaiah

EF 471898

ii) To develop, provide, undertake, design, import, export, distribute and deal in Systems and application software for microprocessor based information systems, off shore software development projects, internet service provider, and solutions in all areas of application including those in Emerging niche segments like Internet and Intranet website applications solutions software enterprise, resource planning, e-commerce, value added products and other business applications either for its own use for sale in India or for export outside India and to design and develop such systems and application software for and on behalf of manufacturers owners and users of computer, telecom, digital, electronic equipments in India or elsewhere in the world.

The Capital contribution of the firm shall be Rs. 1000/- (Rupees One Thousand only) which shall be contributed by the partners in the following proportions.

First party 50 % i.e., Rs. 500 (Rupees Five Hundred Only)

Second party 50 % i.e., Rs. 500 (Rupees Five Hundred Only)

The further Contribution if any required by the firm shall be brought in by the partners in their profit sharing ratio or as may be mutually decided by all partners.

H. M. S. S. S.

6. That the partners shall share the Profit or Loss of the firm as under:

- | | |
|---|-----|
| i) MOHAMMED ANAS H. (the party of the First part) | 50% |
| ii) GAYATHRI V C. (the party of the Second part) | 50% |

7. That the partnership shall be partnership 'AT WILL'.

8. That the partners shall be entitled to simple interest @ 12% per annum or at such other rate but subject to provisions and ceilings laid down u/s 40 (b) of the Income Tax Act, 1961 as may be mutually decided by the partners from time to time. The interest as above shall be charged on profits of the firm

9. That all the partners shall devote their time and attention in the conduct of the affairs of the firm as the circumstances and the business needs may require.

10. The partners shall be entitled to modify the terms relating to remuneration, interest etc., payable to the partners by executing a supplementary deed and any such deed when executed shall have affect, unless otherwise provided, from the First day of the accounting period in which such supplementary deed is executed and the same shall form part of this deed of partnership.

11. That the Total Remuneration payable to Working Partners shall be worked out as under:

- | | | |
|-----|---|---|
| I. | On the first Rs. 3,00,000/- of Book Profits - | Rs. 150000/- or @ 90% of Book Profits, whichever is more. |
| II. | On the balance of the Book Profit | At the rate of 60% |

For the purpose of this clause, "Book Profit" shall mean the net profit, as shown in the Profit and Loss Account for the relevant accounting year, computed in the manner laid down in the chapter IV - D of the Income Tax Act, 1961 as increased by the aggregate amount of the remuneration paid or payable to all the partners of the firm if such has been deducted while computing the net profit.

The remuneration payable to the working partners as above shall be credited to their respective Capital accounts on ascertainment of the book profits at the close of every accounting year. The working partners shall be entitled to draw amounts from time to time against the remuneration payable to them and if their total withdrawals during the year towards their remuneration exceeds the total amount found payable to them on that account, such excess shall be adjusted in their capital account at the close of the accounting year.

12. The Partners covenant and agree that, during the term of their Partnership with "MS WEIZENITH SOLUTIONS" and for three (3) years after the termination thereof, regardless of the reason for the termination, the outgoing partner will not, directly or indirectly, anywhere in the Indian Territory, not set up any Competitive Business solely or together with any person(s). However, if the terminated partner take only prior consent from the partnership firm then it is sufficient to start new business in the same domain.

14/11/2022
V. S. S. S. S. S.

13. The Net Profit of the partnership business as per the accounts maintained by the partnership after deduction of all expenses relating to business activities of the partnership including rent, salaries and other establishment expenses as well as interest and remuneration payable to the partners in accordance with this deed of partnership or any supplementary deed as may be executed by the partners shall be divided and distributed.
14. That none of the parties shall take any loan, stand surety nor give guarantee for any other party or parties on behalf of the firm without having consent of the other partner in writing except for bank loan purpose for carrying on the business of the partnership.
15. That the partners shall be entitled to open bank account/accounts with such bank/banks as mutually decided and the party of the First part and Second part shall be authorized to operate such account/accounts jointly or severally.
16. That the books of accounts shall be closed on 31st March, or on such other date or dates as may be mutually agreed upon, and the net profit and/or losses of the firm shall be divided between the partners after meeting all the business expenses including Depreciation and Bad Debt etc.
17. That the books of accounts of the firm shall be regularly maintained and kept at the principal place of business or any other place as mutually agreed upon by the partners and the same shall not be removed without the permission of the partners and that the partner shall have access to the books of account at any time.

ADMISSION OF NEW PARTNER

18. No Person may be introduced as a new partner without the consent of all the existing partners. Such incoming partner shall give his prior consent to act as Partner of the "firm".
19. The Contribution of the partner may be tangible, intangible, Moveable or immovable property and the incoming partner shall bring minimum contribution as mutually agreed upon by existing partners.
20. The Profit sharing ratio of the incoming partner will be in proportion to his contribution towards "Firm".

RIGHTS OF PARTNER

21. All the partners hereto shall have the rights, title and interest in all the assets and properties in the said "firm" in the proportion of their Contribution.
22. Every partner has a right to have access to and to inspect and copy any books of the "firm".
23. Each of the parties hereto shall be entitled to carry on their own, separate and independent business as hitherto they might be doing or they may hereafter do as they deem fit and proper and other partners and the "firm" shall have no objection thereto provided that the said partner has intimated the said fact to the "firm" before the start of the independent business and moreover he shall not use the name of the "firm" to carry on the said business.

25. "Firm" shall have perpetual succession, death, retirement or insolvency of any partner shall not dissolve the "Firm".

26. On retirement of a partner, the retiring partner shall be entitled to full payment in respect of all his rights, title and interest in the partner as herein provided. However, upon insolvency of a partner his or her rights, title and interest in the "Firm" shall come to an end upon the death of any of the partners herein any one of his or her heirs will be admitted as a partner of the "Firm" in place of such deceased partner. The heirs, executors and administrators of such deceased partners shall be entitled to and shall be paid the full payment in respect of the right, title and interest of such deceased partner.

27. On the death of any partner, if his or her heir opts not to become the partner, the surviving partners shall have the option to purchase the contribution of the deceased partner in the "Firm".

DUTIES OF PARTNERS

28. Every partner shall account to the liability of partnership for any benefit derived by him without the consent of the partners from any transaction concerning the partnership, or from any use by him of the property, name or any business connection of the "Firm".

29. Every partner shall indemnify the "Firm" and the other existing partner for any loss caused to it by his fraud in the conduct of the business of the "Firm".

30. Each partner shall render true accounts and full information of all things affecting the "Firm" to any partner or his legal representatives.

31. In case any of the Partners of the "Firm" desires to transfer or assign his interest or shares in the "Firm" he has to offer the same to the remaining partners by giving 15 days notice. In the absence of any communication by the remaining partners the concerned partner can transfer or assign his share in the market.

32. No partner shall without the written consent of the "Firm" -

I. Employ any money, goods or effects of the "Firm" or pledge the credit thereof except in the ordinary course of business and upon the account or for the benefit of the "Firm".

II. Lend money or give credit on behalf of the "Firm" or to have any dealings with any persons, company or firm whom the other partner previously in writing have forbidden it to trust or deal with. Any loss incurred through any breach of provisions shall be made good with the "Firm" by the partner incurring the same.

III. Enter into any bond or becomes surety or security with or for any person or do knowingly cause or suffer to be done anything whereby the "Firm" property or any part thereof may be seized.

IV. Assign, mortgage or charge his or her share" in the "Firm" or any asset or property thereof or make any other person a partner therein.

V. Compromise or compound or (except upon payment in full) release or discharge any debt due to the "Firm" except upon the written consent given by the other partner.

#N/A
12/10/2019

vi. All the Partners shall be responsible for the doing of all acts, matters and things as are required to be done by "Firm" in respect of compliance of the provisions of this Act including filing of any document, return, statement and the like report pursuant to the applicable laws.

VII. All the Partners shall be responsible for the doing of all acts arising out of this agreement.

VIII. The "Firm" shall indemnify and defend its partners and other officers from and against any and all liability in connection with claims, actions and proceedings (regardless of the outcome), judgment, loss or settlement thereof, whether civil or criminal, arising out of or resulting from their respective performances as partners and officers of the "Firm", except for the gross negligence or willful misconduct of the partner or officer seeking indemnification.

MEETING

32. All the matters related to the "Firm" as mentioned in this agreement shall be decided by a resolution passed by a majority in number of the partners, and for this purpose, each partner shall have one vote.

33. The meeting of the Partners may be called by sending 15 days prior notice to all the partners at their residential address or by mail at the Email ids provided by the individual Partners in written to the "Firm". In case any partner is a foreign resident the meeting may be conducted by serving 15 days prior notice through email. Provided the meeting be called at shorter notice, if majority of the partners agrees in writing to the same either before or after the meeting.

34 The meeting of Partners shall ordinarily be held at the registered office of the Firm or at any other place as per the convenience of partners.

35. With the written Consent of all the partners, a meeting of the Partners may be conducted through Teleconferencing.

*6. Every partner shall ensure that decisions taken by them are recorded in the minutes within thirty days of taking such decisions and are kept and maintained at the registered office of the "Firm".

37. Each partner shall--

(i) Punctually pay and discharge the separate debts and engagement and indemnify the other partners and the "firm" assets against the same and all proceedings, costs, claims and demands in respect thereof.

d) Each of the partners shall give time and attention as may be required for the fulfilment of the objectives of the "firm" business and they all shall be the working partners

[illegible]

CESSATION OF EXISTING PARTNERS

38. Partner may cease to be partner of the "Firm" by giving a notice in writing of not less than thirty days to the other partners of his intention to resign as partner.
39. No majority of Partners can expel any partner except in the situation where any partner has been found guilty of carrying of activity/business of "Firm" with fraudulent purpose.

40. The "Firm" can be wound up with the consent of all the partners subject to the provisions applicable laws.

EXTENT OF LIABILITY OF FIRM

41. "Firm" is not bound by anything done by a partner in dealing with a person if—
- 3) The partner in fact has no authority to act for the "Firm" in doing a particular act; and
- 4) The person knows that he has no authority or does not know or believe him to be a partner of the "Firm".

MISCELLANEOUS PROVISIONS

42. "Firm" shall indemnify each partner in respect of payments made and personal liabilities incurred by him—
- 1) in the ordinary and proper conduct of the business of the Firm; or
- 2) in or about anything necessarily done for the preservation of the business or property of the Firm.

43. The books of accounts of the firm shall be kept at the registered office of the "Firm" for the reference of all the partners.

44. The accounting year of the "Firm" shall be from 1st April of the year to 31st March of subsequent year. The first accounting year shall be from the date of commencement of this "Firm" till 31st March of the subsequent year.

45. The books of accounts of the "Firm" shall be operated by the First and Second Party jointly or severally as mutually agreed.

46. All disputes between the partners of between the Partner and the "Firm" arising out of the Partnership Agreement which cannot be resolved in terms of this Agreement shall be referred for arbitration as per the provisions of the Arbitration and Conciliation Act, 1996 (26 of 1996).

47. This the firm shall not be dissolved in case of death or retirement of any partner and shall continue with remaining partners or with their heirs.

48. That in case of dispute or difference of opinion in the matters of partnership between the partners or any one of the legal representative heirs of the partners, whether during or after the partnership the same shall be referred to the arbitration in the accordance with the statutory modifications for the time being in force and the decision of the arbitrator/arbitrators as appointment shall be accepted by all the partners.

1. In WITNESSES whereof the above parties have signed and sealed this day, 2nd day of AUGUST 2024 mentioned here under.

WITNESSES:

1.

[Signature]
S. S. S. S. S.

2.

[Signature]
S. S. S. S. S.

[Signature]
MOHAMMED ANAS H.
(The party of the First part)

[Signature]
GAYATHRI V C
(The party of the Second part)

[Signature]
R. SHYAMALA, B.A., M.L.
ADVOCATE & NOTARY PUBLIC
NEW NO. 15, 11A STREET,
VYTHIRI, ANNAMALAI STREET,
MYLAPORE, CHENNAI - 600 004.
PHONE: 944 24640127 / 94444134454



Student won prize in Technical event paper presentation



Students won in Hackathon

Learning Programs

These findings have important implications for the design of future research on the effects of social support on health outcomes. First, the results suggest that the effects of social support may be more complex than previously assumed, with different types of support having different effects on different outcomes. Second, the results suggest that the effects of social support may be more context-specific than previously assumed, with the effects of support varying depending on the specific situation or event. Third, the results suggest that the effects of social support may be more long-term than previously assumed, with the effects of support persisting over time. Finally, the results suggest that the effects of social support may be more individual-specific than previously assumed, with the effects of support varying depending on the individual's characteristics and needs.

© 2006 Blackwell Publishing Ltd

• *Veronica* sp.

Left fly: 100% at water entry. >

Spoken results: Subsequent to

Received 10 July 2020

S.A.Engineering College on Coursera

10

Learnings Invited

241

1000

Learn more today

241

100

forested landscape

207

1000

Available online 10 May 2010